



Meen Patel

Nürnberg, Bayern, Deutschland

About

M.Sc. candidate in Autonomy Technologies at Friedrich Alexander Universität Erlangen Nürnberg with hands-on experience in embedded firmware development, ESP32-based AI systems, and drone vision modules. Experienced in real-time systems, microcontroller programming, and IoT integration.

Skills

Programming Languages: C, Embedded C, Python, Assembly (MASM), MATLAB
Embedded Platforms: ESP32, STM32, 8051, Arduino, ARM Cortex-M, DMA, Timers, PWM, ADC, DAC, PWM/Timer-focused testing
Protocols: UART, SPI, I2C, MQTT
RTOS & Firmware: FreeRTOS, Zephyr, State Machines, Interrupt Handling, Task Scheduling, Software Debugging, Firmware-Hardware Integration
Tools: Git/GitHub, Keil, Proteus, ModelSim, Espressif IDE, MATLAB, Linux Platform
Hardware: Oscilloscope, Multimeter, Sensor Integration, Sensor Calibration, Prototype Assembly, JTAG/SWD, ST-Link, Segger J-Link
Computer Vision: OpenCV, Image Processing
Domains: Embedded Systems, IoT, Industrial Automation, Digital Electronics, Robotics, Software Development
Robotics & Autonomy: Sensor Fusion, ROS2, Autonomous Navigation, Path Planning, Computer Vision

Languages

English - B2
German - A1
Gujarati - Native
Hindi - Native

Contact

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📄 github.com/MeenPatel

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg <i>M.Sc. in Autonomy Technologies</i>	2026 – Present <i>Erlangen, Germany</i>
Nirma University <i>B.Tech in ECE</i>	2021 – 2025 <i>Ahmedabad, Gujarat</i>
Vishwabharati English Medium School <i>12th</i>	2020 – 2021 <i>Ahmedabad, Gujarat</i>

Experience

NXON AI Pvt. Ltd. <i>Microcontrollers, espressif</i> <i>Junior Embedded Engineer</i> <i>Embedded Software Developer Intern</i>	Dec. 2024 – Jan. 2026 <i>April 2025 – Jan. 2026</i> <i>Dec. 2024 – April 2025</i>
- Designed and developed a secure lock system using nRF52840 with NFC and BLE connectivity with proper authentication. - Assigned key responsibilities included customizing and enhancing open-source firmware based on specific client requirements and NRF boards, encompassing user interaction features and system performance optimization. - Developed a Gemini-powered project on ESP32 with speaker and microphone integration. - Contributed to the development of a drone vision module, assisting in camera integration and real-time processing under senior engineering supervision.	
eInfochips - An ARROW Company <i>Research</i> <i>R & D Intern</i>	Jun. 2024 – Jul. 2024
Conducted research on AI applications in embedded electronics and wireless IoT systems, analyzing product architecture and connectivity solutions.	
Midas Metcons Pvt. Ltd. <i>Embedded C, 8051</i> <i>Embedded Intern</i>	Jun. 2023 – Dec. 2023
Developed Embedded C firmware for microcontroller-based automated storage systems and performed simulation-based validation.	

Projects

Secure Lock System using nRF52840
Implemented low-latency speech processing with cloud API integration and optimized firmware for stable real-time interaction via NFC and BLE.
ESP32 AI Voice Assistant
Implemented low-latency speech processing with cloud API integration and optimized firmware for stable real-time interaction.
Vertical Lift Module (Warehouse Automation)
Designed RTOS-based task scheduling system for efficient actuator control and IoT-based remote diagnostics.
Vision Control for Drone System
Calibrate camera to correct distortion and detect marker, estimate pose, and calculate movement.
MIPS Assembler in Python
Implemented a script that translates assembly instructions into machine code with error handling to demonstrate assembler functionality.
Image Modulation Using Sound (MATLAB)
Dynamically changed image based on sound frequency and amplitude.

**For more projects, visit my portfolio: meenpatel.github.io/portfolio **

Achievements

AMD Pervasive AI Developer Contest <i>Neuro Glove: Smart Rehabilitation Assistant-AMD provided KR260</i>	Feb 2024 – Aug 2024 <i>Hackster.io</i>
Smart India Hackathon <i>Regional Level Finalist</i>	Sep 2023 <i>E-Cell, Nirma University</i>